



Non-invasive prediction of the secondary enucleation risk in uveal melanoma based on pretreatment CT and MRI prior to stereotactic radiotherapy

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Abstract

Purpose The aim of this study was to develop a radiomic model to non-invasively predict the risk of secondary enucleation (SE) in patients with uveal melanoma (UM) prior to stereotactic radiotherapy using pretreatment computed tomography (CT) and magnetic resonance (MR) images.

Materials and methods This retrospective study encompasses a cohort of 308 patients diagnosed with UM who underwent stereotactic radiosurgery (SRS) or fractionated stereotactic radiotherapy (FSRT) using the CyberKnife system (Accuray, Sunnyvale, CA, USA) between 2007 and 2018. Each patient received comprehensive ophthalmologic evaluations, including assessment of visual acuity, anterior segment examination, fundus examination, and ultrasonography. All patients were followed up for a minimum of 5 years. The cohort was composed of 65 patients who underwent SE (SE+) and 243 who did not (SE−). Radiomic features were extracted from pretreatment CT and MR images. To develop a robust predictive model, four different machine learning algorithms were evaluated using these features.

Results The stacking model utilizing CT+MR radiomic features achieved the highest predictive performance, with an area under the curve (AUC) of 0.90, accuracy of 0.86, sensitivity of 0.81, and specificity of 0.90. The feature of robust mean absolute deviation derived from the Laplacian-of-Gaussian-filtered MR images was identified as the most significant predictor, demonstrating a statistically significant difference between SE+ and SE− cases ($p=0.005$).

Conclusion Radiomic analysis of pretreatment CT and MR images can non-invasively predict the risk of SE in UM patients undergoing SRS/FSRT. The combined CT+MR radiomic model may inform more personalized therapeutic decisions, thereby reducing unnecessary radiation exposure and potentially improving patient outcomes.

Keywords Radiomics · Side effect prediction · Predictive model · Machine learning in medicine · Prognosis prediction

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Uveal melanoma (UM) is the most common primary intraocular malignancy in adults. Historically, enucleation was the standard treatment for UM. However, radiation therapy (RT) has gained prominence in recent years due to its potential to preserve the globe and retain vision [1–3]. Enucleation remains necessary in cases of large tumors or when the eye causes intolerable pain or discomfort [4]. Although RT is an alternative to primary enucleation, some patients still require secondary enucleation (SE) following RT. This SE is often caused by tumor progression or adverse effects associated with RT [5, 6].

Radiotherapy (RT) modalities for treatment of uveal melanoma (UM) include brachytherapy, stereotactic radiosurgery (SRS), fractionated stereotactic radiotherapy (FSRT), and proton therapy [7–9]. The incidence of SE fol-

lowing SRS/FSRT can reach up to 38% [8, 10]. In contrast, the incidence of SE is reported to be 15% and 22.6% following brachytherapy and proton therapy, respectively [6, 11]. Previous studies have identified significant risk factors for SE, including increased tumor height and the development of neovascular glaucoma after RT [12, 13]. Non-invasive techniques are frequently employed in the diagnosis of uveal melanoma. These include ophthalmoscopy, ultrasonography, fluorescein angiography, optical coherence tomography, and magnetic resonance imaging (MRI) [14]. Secondary enucleation risk prediction has generally been attempted according to the results of these methods. To a lesser extent, biopsy is used for diagnosis when other diagnostic methods are inconclusive or in ambiguous cases. Notably, Bosch et al. [12] reported that biopsy results contain valuable information for SE risk estimation. However, eye biopsies carry risks, including complications, potential tumor spread, diagnostic reliability issues, and vision loss [15]. In contrast, radiomics could offer a non-invasive and reliable alternative approach. To date, there is no study employing radiomics to predict the SE risk.

Radiomics is an evolving discipline that enables extraction of quantitative features from radiological images, thereby providing predictive insights. By leveraging mathematical and statistical techniques, radiomics facilitates a thorough analysis of medical imaging data, often incorporating machine learning (ML) and advanced image analysis methods [14–18]. This comprehensive approach allows for personalized assessments [17, 19]. We hypothesized that radiomics could predict the risk of secondary effects prior to administering RT in patients with UM.

The objective of this study was to develop a predictive model that could forecast the risk of SE in UM patients undergoing SRS/FSRT. This approach integrates radiomics to achieve a comprehensive understanding of treatment outcomes and equip clinicians with essential tools for making personalized therapeutic decisions.

Materials and methods

Patient cohort

This retrospective study was approved by the institutional ethics committee and performed in accordance with the Declaration of Helsinki. The study is based on data from 308 patients diagnosed with UM who were treated in our department between 2007 and 2018. The diagnosis of UM was confirmed through ophthalmoscopy. Each patient underwent a comprehensive ophthalmologic assessment, including best-corrected visual acuity testing, anterior segment examination, fundus examination, and fundus fluorescein angiography. These assessments aimed to evaluate

the impact of the tumor on vision and ocular structures. Tumor thickness was measured using A- and B-mode ocular ultrasonography, providing accurate quantification of the tumor's dimensions. Prior to treatment, each patient underwent a comprehensive pretreatment staging process. This process included a detailed review of the patient's medical history, physical examination, blood biochemistry analysis, and imaging studies (abdominal CT and/or PET/CT) to detect possible metastasis. This comprehensive staging was critical for evaluating the extent of disease and for planning appropriate treatment strategies.

All patients underwent SRS/FSRT via the CyberKnife system (Accuray, Sunnyvale, CA, USA). The prescribed fractionation schemes were as follows: 60 Gy in 3 fractions ($n=101$), 54 Gy in 3 fractions ($n=69$), 45 Gy in 3 fractions ($n=96$), 20 Gy in 3 fractions ($n=32$), and 20 Gy in a single fraction ($n=10$). Patients with distant metastasis, missing planning MR images, or a follow-up duration of less than 5 years were excluded from the study. Following treatment, patients were examined at the following intervals: every 3 months within the initial 2 years after treatment, followed by twice-yearly examination over the subsequent 3 years and annual examinations thereafter. The data for this study were sourced from a variety of channels, including departmental charts, hospital records, reports from referring physicians, and records from the General Directorate of Population and Citizenship Affairs. The dataset comprises 65 patients who have experienced SE. Patients with SE are denoted as SE+, while those without it are labeled as SE–.

Pretreatment imaging

The CT images were acquired using the Brightspeed CT simulator (GE Medical System, Milwaukee, WI, USA), and MR scans were conducted with an imaging protocol of structural 3D T1-weighted sequences on a 1.5T MRI system (Signa HDx; GE Healthcare, Milwaukee, WI, USA) and a 1.5T Achieva scanner (Philips Healthcare, Amsterdam, Netherlands). All scans followed the same protocol for all patients (see Supplementary Table 1).

The T1-weighted MR images, CT images, and the gross tumor volume (GTV) contours were obtained from the treatment planning system. Figure 1 illustrates examples of CT, GTV, and MR images from three SE+ and SE– patients.

Preprocessing

All image preprocessing and feature extraction calculations were conducted using in-house developed software based on Python (version 3.9.18). The detailed overall workflow is illustrated in Fig. 2. Before radiomic feature extraction, systematic preprocessing steps were applied as follows: as for

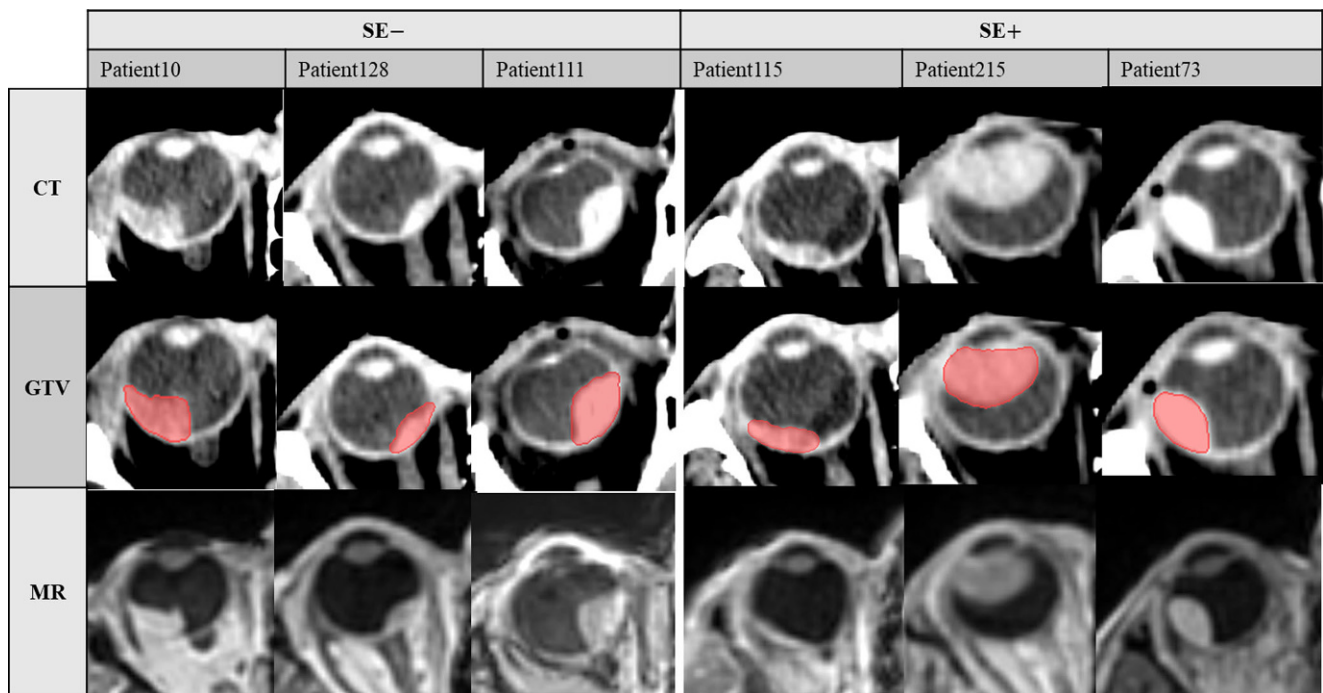


Fig. 1 Treatment planning image samples of patients with (SE+) and without secondary enucleation (SE-) after stereotactic radiotherapy/fractionated stereotactic radiotherapy. *CT* computed tomography, *GTV* gross tumor volume, *MR* magnetic resonance

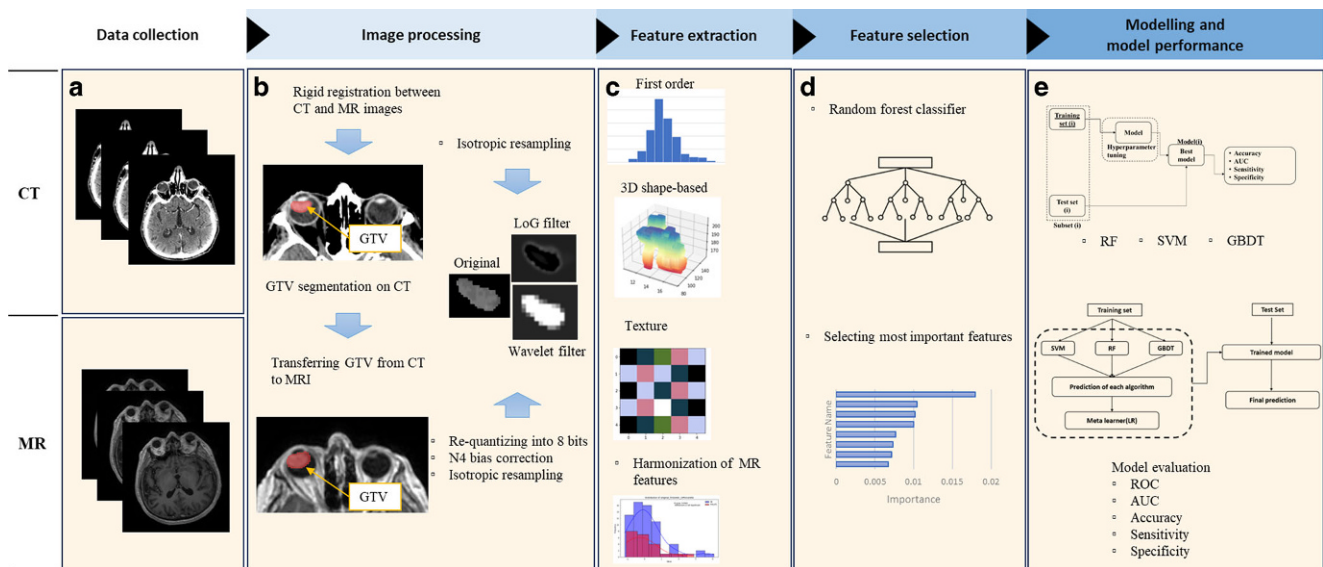


Fig. 2 Study pipeline. **a** Visualization of the collection process for pretreatment computed tomography (CT) and magnetic resonance (MR) images. **b** Illustration of the image processing framework, including the fusion of CT and MR images, gross tumor volume (GTV) delineation, transferring GTVs to MR images, and the application of filters and various image processing techniques. **c** Depiction of the feature extraction and data harmonization for MR features. **d** Using of a random forest (RF) classifier to identify the eight most important features. **e** Representation of the machine learning model development process, including experimental procedures and model comparisons. *SVM* support vector machine, *GBDT* gradient-boosting decision tree, *AUC* area under the curve

CT images, isotropic resampling was performed at a voxel size of $0.8 \times 0.8 \times 0.8 \text{ mm}^3$ using a b-spline interpolation. Regarding MR images, preprocessing was conducted, including re-quantization, N4 bias field correction [20], and isotropic resampling to a voxel size of $0.8 \times 0.8 \times 0.8 \text{ mm}^3$.

Feature extraction

The risk of SE was predicted by employing machine learning models with radiomic features derived from CT and MR images. For this purpose, three types of feature sets were utilized: CT radiomic, MR radiomic, and a combined CT+MR radiomic feature set.

Radiomic features were extracted from the pretreatment images within the GTVs using the PyRadiomics (version 3.1.0) open-source package [21]. The feature extraction encompassed first-order statistical features, shape-based features, and texture features derived from both the original and filtered images. To enhance the features, various filters were applied, including a three-dimensional Laplacian-of-Gaussian (LoG) filter with sigma values of 1, 2, 3, 4, and 5 as well as a wavelet filter. Texture features were computed using several types of matrices: the gray-level co-occurrence matrix (GLCM), gray-level run-length matrix (GLRLM), gray-level size-zone matrix (GLSZM), gray-level dependence matrix (GLDM), and the neighboring gray-tone difference matrix (NGTDM).

After feature extraction, the dataset was divided into training and test sets. The training dataset consisted of 266 patients, while the test dataset comprised 42. The test dataset demonstrated an equal distribution of patients with SE– and SE+. Within the training dataset, only 44 patients were classified as SE+, indicating an inherent class imbalance. To address the imbalance, we implemented an imbalance-adjustment strategy based on the approach described by Schiller et al. [22]. As illustrated in Supplementary Fig. 1A, the original imbalanced training dataset was divided into five balanced subsets. Each subset contained 44 SE+ cases and 44 or 45 randomly selected SE– cases. Subsequently, Z-score normalization was performed to mitigate the potential dominance of features with larger scales in the analysis. This normalization process facilitates the standardization of feature values, thereby ensuring the accurate contribution of each feature to the analysis [23]. Following Z-score normalization, a harmonization technique was applied to the radiomic data derived from MR images. This harmonization process utilized Python's neuroCombat package (version 0.2.12) (neuroCombat; developed by the Department of Biostatistics, Johns Hopkins University, Baltimore, MD, USA), aiming to reduce systematic disparities between radiomic features extracted from different scanners [24]. The harmonization was not applied to CT radiomic features, as only one scanner was used.

Feature selection

Our study employed a feature selection algorithm based on the random forest (RF) algorithm to identify the most pertinent features to find the appropriate input dimensionality for machine learning models, thereby avoiding over- and under-fitting problems and improving the generalization of the developed model. Feature selection with RF improves model accuracy by identifying the most relevant features. Feature importance scores generated by RF indicated the degree of the feature's contribution to the model's predictive power, aiding in understanding and optimizing model performance [25, 26]. Additionally, we optimized the hyperparameters using grid search, which systematically tests model hyperparameters within specified value ranges to determine the best parameters for the optimal model performance [27, 28]. Eight out of the 1316 features were selected for each fold based on their importance scores. These eight features exhibited the highest importance scores, signifying their significant contribution to the model.

Construction and evaluation of predictive models

We constructed four ML classifiers: RF, support vector machine (SVM), gradient-boosting decision tree (GBDT), and a stacking model. The stacking model is an ensemble learning approach that combines RF, SVM, and GBDT as base learners, with logistic regression (LR) serving as the meta-learner. By utilizing the stacking approach, the predictive performance of base learners can be enhanced through the training of a meta-learner. A visual representation of the stacking model architecture is provided in Supplementary Figure A.1. A total of 15 models were used to train the meta-learner.

All computational implementations were carried out in the Python environment. Hyperparameter tuning for each of the five machine learning models—RF, SVM, GBDT, SM, and LR—was performed individually using Bayesian optimization. This approach allowed efficient and model-specific exploration of the hyperparameter space to identify the optimal configuration for each classifier. As described in detail by Wu et al. [29], the training dataset was further divided into inner training and validation subsets using a stratified fivefold cross-validation scheme. Hyperparameter selection was based solely on performance across these internal validation folds. To prevent data leakage and ensure unbiased model evaluation, the independent test dataset was not used at any stage of the model development or tuning process.

Each dataset was divided into five subsets, which were then used sequentially for model training, and subsequently evaluated on an independent test cohort (see Supplementary Fig. 1B).

Table 1 Summary of patient characteristics

Characteristic	All cases	Training cohort	Test cohort	<i>p</i> -value (method)
<i>Number of cases</i>	308	266	42	–
<i>Age (years)</i>	18–85 (median: 56)	18–85 (median: 56)	18–78 (median: 53)	0.25 (Mann–Whitney U test)
<i>Sex</i>				0.71 (chi-square)
Male	164	140	24	–
Female	144	126	18	–
<i>Tumor diameter (mm)</i>	2–23 (median: 9)	2–23 (median: 9)	3–18 (median: 9)	0.31 (Mann–Whitney U test)
<i>COMS classification</i>				0.67 (chi-square)
Small	30	26	4	–
Medium	202	177	25	–
Large	76	63	13	–
<i>T stage</i>				0.25 (chi-square)
T1	68	60	8	–
T2	124	110	14	–
T3	94	79	15	–
T4	22	17	5	–
<i>Treatment dose</i>				
BED _{3Gy} (Gy)	43–550 (median: 460)	43–482 (median: 460)	67–550 (median: 378)	0.07 (Mann–Whitney U test)
BED _{10Gy} (Gy)	20–211 (median: 180)	20–188 (median: 180)	38–211 (median: 151)	0.07 (Mann–Whitney U test)

COMS collaborative ocular melanoma study: large > 10 mm in apical height and > 16 mm in basal diameter, medium 2.5–10 mm in apical height and ≤ 16 mm in basal diameter, small 1.5–2.4 mm in apical height and 5–16 mm in basal diameter; BED biologically effective dose

To assess the predictive capability of the classifiers, various performance metrics were computed. The metrics included the area under the receiver operating characteristic curve (AUC), accuracy, sensitivity, and specificity. These metrics provided the models' effectiveness in distinguishing between SE+ and SE– cases across the different datasets.

Statistical analysis

Statistical analyses were performed using SPSS version 23 software (IBM Corp., Armonk, NY, USA). For the demographic distributions of patients (Table 1), Mann–Whitney U tests were employed to assess significant differences for continuous variables, which did not follow the normal distribution. Chi-squared tests were used for the other categorical variables. As a result of the tests, $p < 0.05$ was considered statistically significant.

Results

The clinical characteristics of the 308 patients included in this study are detailed in Table 1, which also shows their distribution across both the training and test datasets. The training dataset consisted of 266 cases, with 44 SE+ and 222 SE– patients. The test dataset included 42 cases, with

an equal number of patients in each category (21 SE+ and 21 SE–). The results of the statistical analysis demonstrated that there were no significant differences between the training and test datasets.

A total of 65 patients (21%) underwent SE during the follow-up period. Among the cases, 38 (58%) underwent SE due to tumor progression, while 27 (42%) underwent SE due to side effects related to RT. The SE rate was 13% for the small tumor group, 18% for the medium tumor group, and 29% for the large tumor group. Regarding tumor location, the SE rate was 47% for anteriorly located tumors, 22% for tumors located in the middle region, and 15% for posteriorly located tumors.

A total of 12 distinct predictive models were developed, each utilizing a different feature set (CT, MR, and CT + MR) and a distinct ML algorithm (RF, SVM, GBDT, and SM). Table 2 presents the performance metrics for each model, including accuracy, sensitivity, specificity, and AUC. The test AUC for RF, SVM, GBDT, and stacking models with the CT image features were 0.52, 0.57, 0.54, and 0.80, respectively. The test AUC values for RF, SVM, GBDT, and stacking models with the MR image features were 0.78, 0.75, 0.79, and 0.83, respectively. The test AUC values for RF, SVM, GBDT, and stacking models with CT + MR combined features were 0.74, 0.68, 0.69, and 0.90, respectively. The stacking model achieved the highest AUC among the

Table 2 Sensitivity, specificity, accuracy, and area under the receiver operating characteristic curve results of the studied models

Metric		RF		SVM		GBDT		Stacking model	
		Training	Test	Training	Test	Training	Test	Training	Test
CT	AUC	0.76	0.52	0.84	0.57	1.00	0.54	1.00	0.80
	Accuracy	0.75	0.55	0.93	0.60	1.00	0.56	1.00	0.71
	Sensitivity	0.80	0.50	0.95	0.52	1.00	0.50	1.00	0.81
	Specificity	0.70	0.53	0.91	0.62	1.00	0.59	1.00	0.62
MR	AUC	0.97	0.78	1.00	0.75	1.00	0.79	1.00	0.83
	Accuracy	0.97	0.82	1.00	0.82	1.00	0.84	1.00	0.87
	Sensitivity	0.98	0.80	1.00	0.80	1.00	0.81	1.00	0.76
	Specificity	0.95	0.75	1.00	0.66	1.00	0.76	1.00	0.90
CT+	AUC	0.81	0.74	0.96	0.68	1.00	0.69	1.00	0.90
MR	Accuracy	0.81	0.79	0.88	0.76	1.00	0.74	1.00	0.86
	Sensitivity	0.89	0.77	0.84	0.71	1.00	0.69	1.00	0.81
	Specificity	0.73	0.71	0.91	0.66	1.00	0.68	1.00	0.90

CT computed tomography, MR magnetic resonance, AUC area under the curve, RF random forest, SVM support vector machine, GBDT gradient-boosting decision tree

models with the three feature types (CT, MR, and CT+MR). The combined CT+MR image features enabled the stacking model to reach the highest AUC (0.90), with an accuracy of 0.86, sensitivity of 0.81, and specificity of 0.90 (Fig. 3).

A total of 1316 features were extracted from each image type of CT and MR for each patient. These extracted features were subsequently utilized to create the CT+MR feature set. The eight features with the highest importance scores for each set (CT, MR, and CT+MR) are presented in Supplementary Fig. 2. For CT, MR, and CT+MR, the most significant features were identified as wavelet-HHL_glcml, wavelet-HHH_glszm_ZoneEntropy, and log-sigma-5-0-mm-3D_firstorder_RobustMeanAbsoluteDeviation, respectively. Figure 4 presents violin plots illustrating the distribution of these features for SE- and SE+ patients. The difference in the distribution of the wavelet-HHL_glcml feature scores, which have the highest importance score for CT images, is not statistically significant between SE- and SE+ patients ($p=0.457$). However, the wavelet-HHH_glszm_ZoneEntropy feature scores from MR images demonstrate a statistically significant difference between SE- and SE+ patients ($p=0.026$). Furthermore, the log-sigma-5-0-mm-3D_firstorder_RobustMeanAbsoluteDeviation feature from the combined CT+MR set also demonstrates a statistically significant difference between SE- and SE+ patients, with the lowest p -value among the features ($p=0.005$).

Discussion

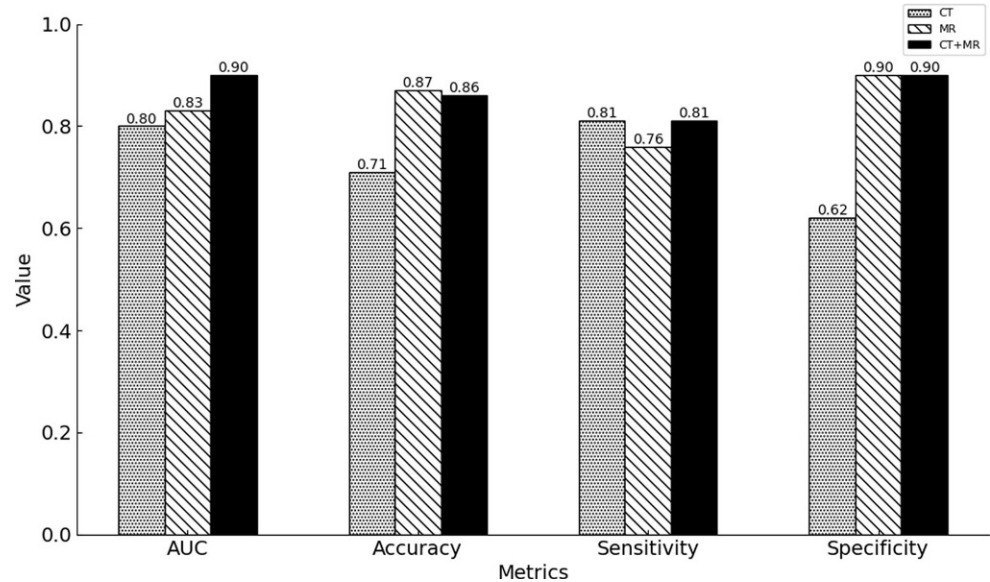
This study aimed to investigate the potential of radiomic features derived from pretreatment CT and MR images to

predict the risk of SE in patients with UM undergoing SRS/FSRT. The findings demonstrated that radiomic predictive models, particularly the stacking model applying to CT+MR features, exhibited satisfactory performance in classifying patients who would need SE following RT. The highest AUC of 0.90 in the test cohort was observed with the CT+MR feature set using the stacking model, indicating that this model exhibited superior predictive capability.

The robust mean absolute deviation (RMAD) feature for the Laplacian-of-Gaussian (LoG)-filtered MR images emerged as the most significant feature in the CT+MR model. The LoG filter enhances and detects edges within the image by first applying Gaussian smoothing to reduce noise, followed by the Laplacian operator to accentuate edge information. RMAD quantifies the variability in voxel intensities by calculating the mean absolute difference between each voxel's intensity and the median intensity of the entire image or region of interest (ROI). In our analysis, RMAD demonstrated the lowest p -value in discriminating between SE- and SE+ cases, indicating a statistically significant distinction between these groups. This indicates that RMAD is capable of capturing critical variability in intensity patterns, which may provide valuable insights into the tumor's characteristics or its response to treatment. The significance of RMAD in the CT+MR model highlights its potential to enhance the predictive power of radiomic features in clinical settings.

It is evident that accurately predicting the risk of SE prior to administering SRS/FSRT offers substantial clinical benefits. The SRS/FSRT process involves numerous preparatory steps, such as immobilizing the eye before each treatment, acquiring treatment images, contouring, and planning [30, 31]. In the event that SE becomes necessary after SRS/FSRT, the initial choice of RT may be unwarranted, as it

Fig. 3 Comparison of performance metrics for the test cohort using the stacking model (SM) across different feature sets derived from computed tomography (CT), magnetic resonance (MR), and combined CT + MR



results in a waste of clinical resources and exposes patients to potential RT complications [32–34]. Anticipating the risk of SE may prevent unnecessary radiation exposure and thus reduce the risks associated with secondary cancer development and RT-related complications. Furthermore, performing SE after RT presents technical challenges and is associated with a higher incidence of complications compared to primary enucleation, as reported by Chau et al. [35]. Consequently, our predictive model, which exhibits a high AUC of 0.90 for prediction of the risk of SE, may significantly mitigate such risks by facilitating more informed decision-making. The proposed model has the potential to

enable clinicians to more accurately identify patients who will benefit from RT, thereby improving clinical workflows and avoiding unnecessary treatments.

In our study, the SE rate was 21% among 310 patients. In comparison, Bosch et al. [12] reported an SE rate of 16% in 118 patients treated with FSRT. Similarly, Fabian et al. [13] observed a 19% SE rate among 515 patients treated with plaque RT, proton therapy, and transpupillary thermotherapy, while Wang et al. [6] observed a 15% SE rate in 880 patients undergoing plaque RT. Regarding the reasons for SE, Bosch et al. [12] indicated that 31% of SE cases were due to treatment failure, with the remaining

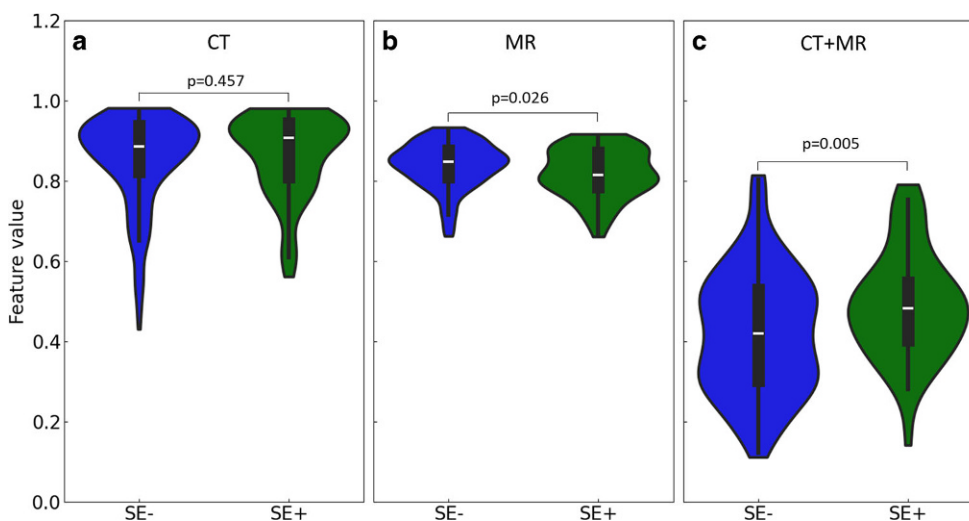


Fig. 4 Violin plots illustrating the distribution of radiomic features with the highest importance scores, distinguishing between patients with (SE+) and without (SE-) secondary enucleation after stereotactic radiosurgery/fractionated stereotactic radiotherapy. **a** Shows wavelet-HHL_glm_Imc1, **b** represents wavelet-HHH_glszm_ZoneEntropy, and **c** demonstrates log-sigma-5-0-mm-3D_firstorder_RobustMeanAbsoluteDeviation (MR) for the computed tomography (CT), magnetic resonance (MR), and CT + MR sets, respectively. Blue represents patients without secondary enucleation, while green represents patients with secondary enucleation. Min-max normalization was performed to ensure all data were displayed on the same scale

cases attributed to complications. In contrast, Wang et al. [6] found that 39.4% of SE cases resulted from complications and 46.9% from treatment failure. Fabian et al. [13] reported that 21% of SE cases were due to complications, with the remaining 79% attributed to treatment failure. In our study, treatment failure emerged as the predominant cause of SE, accounting for 58% of SE cases. With respect to risk factors for SE, Bosch et al. [12] observed that a large tumor height was associated with an increased risk of SE, highlighting high mitotic count and the gain of chromosome 8q as common features in SE cases. Wang et al. [6] identified neovascular glaucoma as the primary cause of SE, whereas Fabian et al. [13] emphasized a high-risk group exhibiting radioresistance to treatment. These findings underscore the fact that while treatment failure and complications are the main contributors to SE, their specific rates vary across studies [6, 12, 13, 36]. Moreover, the clinical delineation of SE risk factors remains incomplete. Our study contributes to the existing literature by validating the role of radiomics in predicting the risk of SE, thereby enhancing our understanding of its predictors.

The SE rate varied notably with both tumor size and location. Patients with large tumors exhibited the highest SE rate (29%), followed by those with medium-sized (18%) and small tumors (13%), indicating that tumor volume is a significant prognostic factor. This aligns with the findings of Sreenivasa et al. [37], who reported that tumor volume was the most robust physical prognostic marker for local control and disease progression. In our study, tumors located in the anterior segment of the eye were associated with a substantially higher SE rate (47%) compared to those situated in the middle (22%) or posterior (15%) regions. These results support the notion that anteriorly located and larger tumors may be more susceptible to treatment-related complications or less responsive to radiotherapy, ultimately increasing the likelihood of SE.

While the SE rate in our study cohort was 21%, the current standard SE rate in centers with more advanced treatment techniques may be below 10% [37]. In such a scenario, the clinical utility of our prediction model warrants reevaluation. A lower incidence of SE would reduce the absolute number of events, potentially limiting the statistical power of predictive models. However, the ability to identify rare but high-impact adverse outcomes becomes even more valuable in a context where complications are infrequent. Accurate prediction of the SE risk, even in a low-incidence setting, could help clinicians to avoid unnecessary radiotherapy in patients with a high likelihood of eventual enucleation. Furthermore, the model may assist in tailoring follow-up strategies for high-risk individuals and contribute to individualized treatment decision-making. Importantly, our model could also serve as a complementary tool for assessing the effectiveness of emerging treatment modalities

by identifying patients most likely to benefit. Therefore, even with decreasing SE rates, the predictive model retains its potential clinical relevance, particularly for optimizing patient selection and minimizing overtreatment.

It is important to acknowledge the limitations of our study. Firstly, as the research was conducted within a single department, the findings may be influenced by specific institutional practices, equipment, and patient demographics. This context-specific nature may constrain the generalizability of our results to other clinical settings or broader populations. Secondly, the dataset exhibits a notable imbalance, with only 65 cases of SE+ compared to 243 cases with SE-. This imbalance presents challenges for model training and performance, as the models may be biased towards the majority class, i.e., SE-. Although imbalance-adjustment strategies were implemented, the relatively low number of SE+ cases could impact on the robustness and generalizability of our predictive models, potentially leading to overfitting or under-representation of SE+ cases. Furthermore, while all CT scans were acquired using the same machine, MR images were obtained using different scanners, thereby introducing variability. Despite efforts to harmonize radiomic features using neuroCombat, systematic differences due to varying imaging protocols and equipment could still influence feature extraction and, consequently, model performance. The heterogeneity in MR imaging may introduce biases that affect the reliability of the features derived from these images. Additionally, our study did not consider potential variables beyond radiomic features that might influence the risk of SE, such as genetic factors or specific dosimetric parameters. These unaccounted factors could play a significant role in the SE risk and might provide additional predictive value.

Conclusion

This study demonstrates the potential of radiomic features derived from pretreatment CT and MR images to predict the risk of SE in UM patients undergoing SRS/FSRT. Notably, the stacking model, particularly the combined CT+MR model, achieved a high AUC of 0.90, demonstrating superior predictive accuracy compared to traditional clinical models. The RMAD feature, identified as the most significant in distinguishing SE- from SE+ cases, appears to capture critical aspects of tumor characteristics and treatment responses. This highlights its utility in refining patient selection for RT, potentially leading to more targeted and effective treatment strategies. The promising results of our radiomics-based approach represent a substantial advancement in personalized treatment strategies for UM, offering a valuable tool for enhancing clinical decision-making and improving patient outcomes.

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Conflict of interest Y. Yedekci, H. Arimura, Y. Jin, M.T. Yilmaz, T. Kodama, G. Ozyigit, and G. Yazici declare that they have no competing interests.

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